

Memory Studio: Intergenerational Storytelling through Parent-Child Prompt Engineering with Generative AI

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Intergenerational storytelling strengthens family identity, cultural continuity, and relationships, yet distance, language barriers, and the loss of older relatives can disrupt the transmission of family memories. Meanwhile, generative AI offers new opportunities for children to develop AI literacy, although existing learning initiatives rarely address collaboration among children, parents, and grandparents. We present *Memory Studio*, a family-centered platform that helps elementary school children and parents transform grandparents' photo-less memories into AI-generated images. The platform combines a PARTS-based, Mad Libs-style prompting interface with a hands-on prompt board and keyword tiles. Through selecting memories, collaboratively constructing and testing prompts, revising generated images, and gathering grandparents' feedback, families practice prompt engineering while discussing how AI interprets, omits, or fabricates story details. *Memory Studio* positions prompting as a relational and empathetic activity rather than a purely technical skill. This work contributes design principles for supporting intergenerational storytelling, collaborative AI learning, and critical reflection on generative AI in family settings.

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1 Introduction

Intergenerational storytelling helps children preserve family memories, cultural values, and relationships across generations [11]. It can strengthen children's identity, belonging, and resilience by connecting them to a broader family narrative [5, 10]. However, geographic separation, cultural and language barriers, and the loss of older family members increasingly disrupt the transmission of oral histories. Existing memory technologies also tend to emphasize recording and storing stories rather than supporting the collaborative process through which family members question, interpret, and reconstruct memories together [10].

At the same time, generative AI is becoming part of children's creative activities, creating opportunities to learn AI literacy and prompt engineering with collaborators or supporters. Research on child-AI co-creation emphasizes the importance of balancing creative support with children's agency, and family or peer collaboration [1]. Prompt engineering is also increasingly recognized as part of broader AI literacy as it requires users to formulate goals, evaluate outputs, and critically revise their skills that can be supported through instructional scaffolding [8, 13].

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Yet most AI literacy initiatives focus on schools or individual learners, with limited attention to how families can support this learning at home [2, 4]. Although parents often act as collaborators, scaffolders, and teachers during children’s technology use, little research has examined how children, parents, and grandparents can jointly learn prompt engineering through intergenerational storytelling [2]. We therefore propose *Memory Studio*, a family-centered platform that helps children and parents transform grandparents’ memories into AI-generated images. By combining structured prompting with intergenerational storytelling, *Memory Studio* supports both family memory preservation and collaborative AI literacy learning in informal settings.

2 Background

Prior HCI research frames intergenerational storytelling as more than the transfer or preservation of information [7, 9, 10]. Storytelling involves a two-way exchange in which older adults share experiences while younger family members ask questions, interpret details, and connect those experiences to their own understanding [4]. These interactions can support family identity, cultural continuity, and relationships across generations [9, 10]. Studies with immigrant families further show that collaborative storytelling can help bridge differences in language, cultural knowledge, and generational experience, although these same differences may also make communication difficult [10]. This work suggests that storytelling technologies should support conversation, feedback, and shared interpretation rather than treating recorded memories as fixed representations of the past.

Research on family technology learning shows that parents can act as teachers, collaborators, scaffolders, and learners, creating opportunities for conversation, guidance, and shared meaning-making when adults actively participate rather than simply supervise [4, 15]. However, family-centered resources for AI literacy remain limited, and there is still insufficient understanding of how parents and children can jointly explore AI concepts at home [2]. Emerging projects demonstrate that playful and tangible storytelling activities can help children construct prompts, compare generated outcomes, and discuss how AI interprets their inputs [2, 3, 7]. *Memory Studio* builds on these directions by using grandparents’ memories as socially meaningful content through which children and family members can collaboratively practice prompting, evaluate AI-generated representations, and discuss the capabilities and limitations of generative AI.

3 Design Principles

Memory Studio is a family-centered, intergenerational storytelling platform that helps elementary school children transform grandparents’ photo-less memories into AI-generated images while learning prompt engineering. Using a Mad Libs-style¹ format, children and parents complete structured prompts designed based on the PARTS framework recommended by Google for Education [6]: it defines the prompter’s role (Persona), sets the goal (Aim) and the intended recipient (Recipients), describes key scene parameters (Theme), and specifies the output format (Structure). Before using generative AI, children and parents can brainstorm with a hands-on prompt board and keyword tiles, providing family-centered scaffolding for young learners who may have limited language and AI literacy [8, 13]. The board includes ten color-coded blanks where children and parents place matching keyword tiles, encouraging their discussion and collaborative prompt development.

The activity follows six steps. (1) **Motivation**: An elementary student and parents select a grandparent’s memory card, which gives them a shared intergenerational narrative goal. (2) **Brainstorming**: Build a prompt together using a hands-on prompt board and keyword tiles. (3) **Prompt testing**: Test it in Memory Studio. This tool allows prompt building through keyword selection to

¹Inspired by the *Mad Libs* word game, in which players complete a story by filling in predefined blanks with words or phrases.

support users with limited writing or AI literacy. (4) **Prompt improvement**: Revise it through role-switching, discussing and refining the next prompt. (5) **Moving forward** (optional): Students who want more flexibility can copy their *Memory Studio* prompt into a commercial GenAI tool and refine it under parental supervision. (6) **Presentation**: They share the final image and prompt with parents and grandparents, getting feedback from them and reinforcing the sense of intergenerational understanding.

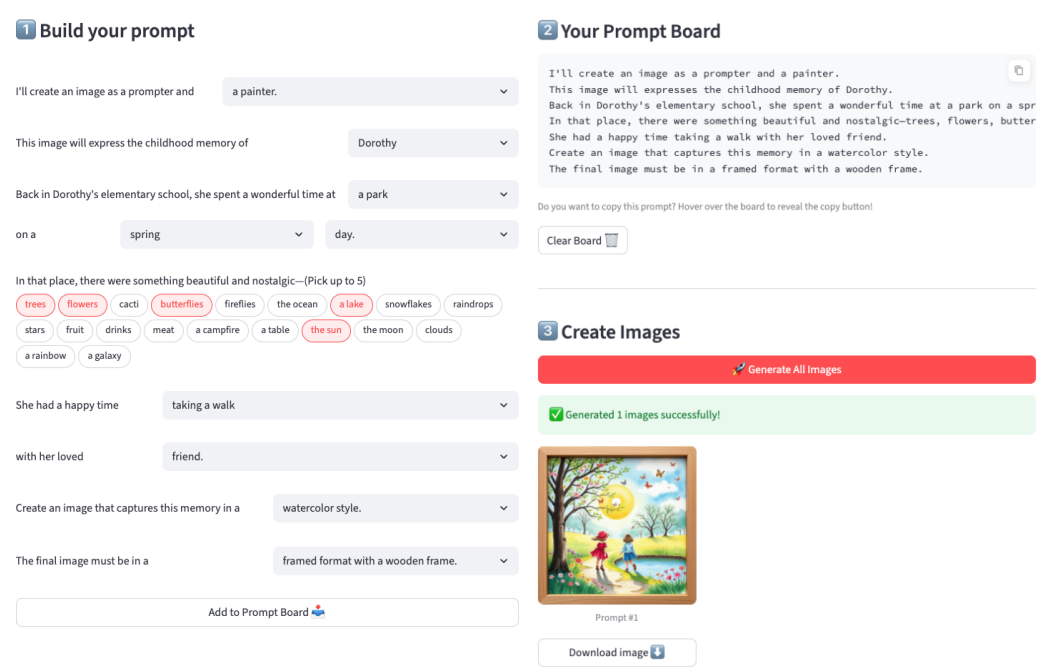


Fig. 1. The screenshot of Memory Studio. This Mad Libs-style tool helps children and parents turn grandparents' memories into AI-generated images together. (<https://sm-prompt.streamlit.app/>)

The following principles synthesize the project introduction, the current prototype, and prior work on intergenerational storytelling and family-centered AI learning.

3.1 Center relational and intergenerational storytelling, not only memory capture

Intergenerational storytelling transforms prompt engineering into a meaningful family activity by preserving grandparents' memories while naturally prompting conversations about AI's creative capabilities, inaccuracies, and limitations. Thus, *Memory Studio* should center relational and intergenerational storytelling by helping children connect with grandparents' lived experiences and refine AI-generated images through their feedback. Rather than simply recording memories, the platform should encourage questions, explanations, and shared interpretation that bridge differences in language, historical knowledge, and generational context [9, 10].

3.2 Use empathy and family relevance as the motivation for learning prompt engineering

By grounding prompt engineering in meaningful family stories, *Memory Studio* gives children a concrete reason to consider which details and emotions matter and whether the generated image

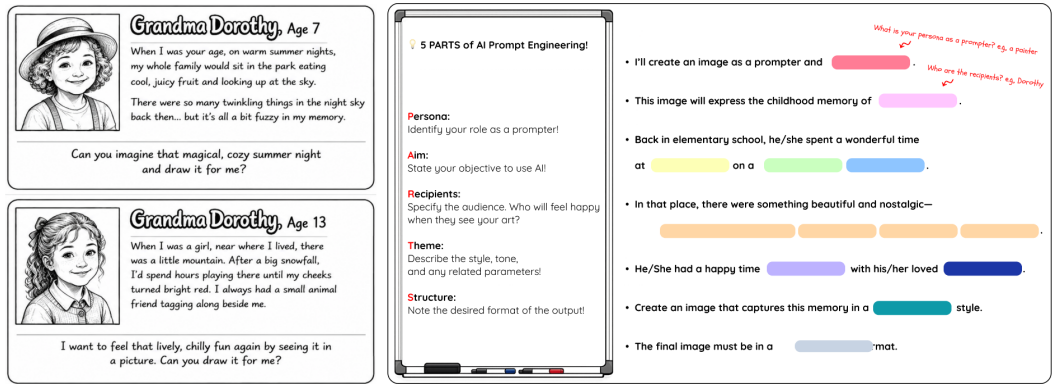


Fig. 2. Hands-on materials support collaborative storytelling before families use the online application. The left shows a grandparent memory card selected by children and parents, while the right shows a PARTS-based prompt board for joint brainstorming.

respectfully represents the grandparent’s memory. This links AI learning with empathy, identity, and intergenerational connection [1, 10].

3.3 Scaffold prompting through family collaboration

Young learners may find it difficult to write complete prompts because the task requires both language literacy and AI literacy. *Memory Studio* should lower this entry barrier through collaboration among children, parents, and grandparents. Structured prompts, keyword tiles, and shared discussion can reduce language and AI literacy barriers, while family members help clarify memories, suggest relevant details, and reflect on how each revision changes the generated image. This approach builds on research showing that family members can learn about AI through shared experimentation and shifting teaching roles [2].

4 Workshop Discussion

This work invites discussion on how future family-centered AI can strengthen intergenerational connections by distributing prompting, interpretation, and feedback across children, parents, and grandparents. In particular, we explore how family-centered AI systems can support intergenerational collaboration and ongoing identity construction while making each family member’s participation and roles visible and negotiable.

We also seek to discuss design considerations for responsible family use and safety, including how family-centered AI systems can help children recognize omissions, fabricated details, and stereotypes in AI outputs [2, 12]. Key discussion points include how to support productive family reflection, select child-appropriate models, minimize personal data collection, and provide clear controls for saving, sharing, and deleting generated content [14].

5 Future Directions

Future work will empirically test and evaluate *Memory Studio* with children, parents, and grandparents in home and informal learning settings. We will examine how families share prompting roles, incorporate feedback from grandparents, and develop intergenerational understanding and critical AI literacy. We also plan to consider diverse contexts and features, such as multilingual storytelling, revision tracking, child-appropriate models, and stronger privacy controls.

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